

Honours Analysis 1 Tutorials (Sec. 003)

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Please note that this document has not been proofread and may contain typos. Feel free to see or email me if you have any questions.

ANNOUNCEMENT: Due to the scheduling of a seminar, I would like to move my Thursday office hours to the 4:00pm-5:00pm slot. Please let me know if this causes any scheduling issues.

1 The Cantor Set

We devote this tutorial to the construction of an infamous topological structure known as the *Cantor set*. It is this set which helped give rise to the subject known today as point-set topology. The Cantor set is very special in the sense that it defies intuition. For one, the Cantor set is a *closed uncountable set* that does not contain **any** open interval. Moreover, the Cantor set is both totally disconnected and nowhere dense (these terms will be defined rigorously in a few moments). Despite this, the Cantor set does not contain any *isolated points* (again, we will give precise meaning to this term shortly). In short, the Cantor set is a prime example of how weak our ingrained topological intuition can be.

1.1 Constructing the Cantor Set

We begin with the closed and bounded interval $[0, 1]$, which has length equal to 1. Let us put $A_0 := [0, 1]$. We can now subdivide A_0 into three **disjoint bounded** intervals, each of length $\frac{1}{3}$, according to the following rule:

$$A_0 := \left[0, \frac{1}{3}\right] \cup \left(\frac{1}{3}, \frac{2}{3}\right) \cup \left[\frac{2}{3}, 1\right].$$

Note that the three intervals given above are pairwise disjoint (i.e. none of these intervals have common points). The idea is now to *remove* what we will call the

middle open third interval. In the case of A_0 , this “middle open third interval” will simply be the open interval

$$\left(\frac{1}{3}, \frac{2}{3}\right).$$

Deleting this interval from A_0 , we obtain a *new* subset of $[0, 1]$ given by

$$A_1 := \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right].$$

Note that A_1 is actually a subset of A_0 . Moreover, since finite unions of closed sets are always closed, A_1 will be both closed and bounded. Now, A_1 consists of two intervals, each having length $\frac{1}{3}$. We will now remove the “middle open third interval” from each of these two sub-intervals. To do this, we decompose $\left[0, \frac{1}{3}\right]$ into thirds as follows:

$$\left[0, \frac{1}{3}\right] = \left[0, \frac{1}{9}\right] \cup \left(\frac{1}{9}, \frac{2}{9}\right) \cup \left[\frac{2}{9}, \frac{1}{3}\right].$$

Thus, when removing the “open middle third” interval from $\left[0, \frac{1}{3}\right]$, we would be left with

$$\left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right]. \quad (1.1)$$

Similarly, we break the second interval in A_1 into thirds:

$$\left[\frac{2}{3}, 1\right] = \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left(\frac{7}{9}, \frac{8}{9}\right) \cup \left[\frac{8}{9}, 1\right].$$

The “open middle third” interval corresponding to the interval above is therefore $\left(\frac{7}{9}, \frac{8}{9}\right)$. Deleting this from $\left[\frac{2}{3}, 1\right]$ leaves us with the following:

$$\left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right]. \quad (1.2)$$

Thus, (1.1)-(1.2) consist of the sets leftover after deleting the “open middle third” intervals from the two subintervals of A_1 , respectively. We then define

$$A_2 := \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right].$$

Again, A_2 is the disjoint union of finitely many closed intervals in $[0, 1]$. Hence, A_2 is closed and bounded. Note that every subinterval in the definition of A_2

has length equal to $\frac{1}{3^2}$. We now apply the last steps recursively. Given A_n , for some $n \in \mathbb{N}$, we remove the “open middle third” interval from every (disjoint) closed subinterval given in the definition of A_n . After removing an “open middle third” interval from a closed interval $[a, b]$, we are always left with two disjoint closed intervals. Therefore, upon removing the “open middle third” subintervals from each of the subintervals of A_n , we will be left with a finite family of disjoint closed intervals, each contained in A_n . We then define A_{n+1} to be the union of these closed intervals. The following properties will hold for this A_{n+1} .

- A_{n+1} is closed. Indeed, A_{n+1} is by definition the finite union of disjoint closed intervals.
- $A_{n+1} \subseteq A_n$. Certainly, every interval in the definition of A_{n+1} was obtained by removing a “middle open third” interval from some subinterval of A_n . In particular, every interval that makes up A_{n+1} is a subset of A_n . Since A_{n+1} is merely the union of these intervals, we get $A_{n+1} \subseteq A_n$.
- $0 \in A_n$ for every $n \in \mathbb{N}$. Indeed, this follows by observing that the end-points of intervals are never removed when deleting the “open middle third” interval. Similarly, $1 \in A_n$.

Therefore, for every $n \geq 0$, we obtain a sequence of closed and bounded sets, denoted by $\{A_n\}_{n \in \mathbb{N}_0}$. Note that, by construction, $0 \in \bigcap_{n \in \mathbb{N}_0} A_n$. In particular, this intersection is non-empty.

Definition 1 (Cantor Set). The Cantor set $\mathfrak{C}$ is the non-empty set

$$\mathfrak{C} := \bigcap_{n \in \mathbb{N}_0} A_n. \quad (1.3)$$

Note that $\mathfrak{C} \subseteq A_0 \subseteq [0, 1]$. Moreover, since A_n is closed for each $n \in \mathbb{N}$, it is immediate from the intersection property of closed sets that $\mathfrak{C}$ is closed.

Since the Cantor set is very difficult to picture, I have included a depiction from Wikipedia below.



Figure 1: Six iterations of the Cantor set

1.2 The Cantor Set is Uncountable

Here, we prove the following result regarding the Cantor Set $\mathfrak{C}$.

Proposition 1. *The Cantor set $\mathfrak{C}$ defined in (1.3) is uncountable.*

Proof. Suppose, by way of contradiction, that $\mathfrak{C}$ is countable and enumerate its elements:

$$\mathfrak{C} := \{x_1, x_2, \dots, x_n, \dots\}.$$

By construction, $\mathfrak{C} = \bigcap_{n \in \mathbb{N}_0} A_n \subseteq A_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right]$. Since these two intervals are disjoint, the point x_1 only belongs to one of these two intervals. Let I_1 denote the interval **not** containing this point x_1 . Dividing this subinterval I_1 by removing its “open middle third” interval, we obtain two disjoint closed subintervals of A_2 (each of length $\frac{1}{3^2}$). Again, since these two intervals are disjoint and $x_2 \in A_2$, we see that x_2 cannot belong to both of these subintervals. Let I_2 be any one of these two subintervals **not** containing x_2 . Since I_2 was obtained by deleting points from I_1 , we clearly have $I_2 \subseteq I_1 \subseteq [0, 1]$.

We proceed recursively, obtaining a nested sequence

$$I_1 \supseteq I_2 \supseteq \dots \supseteq I_n \dots$$

of closed and bounded intervals, with $x_n \notin I_n$ for every $n \in \mathbb{N}$. By the *nested interval property* of $\mathbb{R}$, we know that $\bigcap_{n \in \mathbb{N}} I_n$ is non-empty. Let now $x \in \bigcap_{n \in \mathbb{N}} I_n$ so that $x \in I_n$ for all $n \in \mathbb{N}$. By construction, $I_n \subseteq A_n$ whence $x \in \bigcap_{n \in \mathbb{N}_0} A_n = \mathfrak{C}$. Thus, x belongs to the Cantor set $\mathfrak{C}$ and must be present in our enumeration. On the other hand, $x_n \notin I_n$ for every index n . Since $x \in I_n$, this forces $x \neq x_n$ for all $n \in \mathbb{N}$. We have therefore found an element of $\mathfrak{C}$ not present in our enumeration. Hence $\mathfrak{C}$ cannot be countable. $\square$

1.3 The Topology of $\mathfrak{C}$

We now discuss the surprising topological aspects of the Cantor set. We have already seen that $\mathfrak{C}$ is closed and bounded (making it an example of what is known as a *compact* set). Now, we claim that $\mathfrak{C}$ does not contain any open interval (and hence does not contain any non-trivial interval).

Proposition 2. *The Cantor set $\mathfrak{C}$ does not contain any non-trivial interval.*

Proof. Here we have to look back at our construction of the Cantor set. At the n^{th} stage of the construction, A_n is the *disjoint* union of 2^n closed intervals in $[0, 1]$,

each having length 3^{-n} . Thus, any interval contained in A_n must have length no larger than 3^{-n} . Hence, if I is a non-trivial interval contained in $\mathcal{C}$, it must satisfy

$$|I| \leq \inf_{n \geq 1} 3^{-n}.$$

Here, $|I|$ simply denotes the length of I . By Bernoulli's inequality,

$$3^n > n \quad \text{for all } n \geq 1.$$

Thus,

$$0 \leq \inf_{n \geq 1} 3^{-n} \leq \inf_{n \geq 1} \frac{1}{n} = 0.$$

We then see that $|I| = 0$, which is absurd for any non-trivial interval. We conclude that $\mathcal{C}$ does not contain any non-trivial interval. $\square$

Corollary 1.1. *The Cantor set $\mathcal{C}$ has empty interior. This makes $\mathcal{C}$ an example of a nowhere dense set.*¹

Proof. Let $\mathring{\mathcal{C}}$ denote the interior of $\mathcal{C}$. Then, $\mathring{\mathcal{C}}$ is a subset of $\mathcal{C}$ by definition. If $\mathring{\mathcal{C}}$ were non-empty, then for any point $x \in \mathring{\mathcal{C}}$ we could find $\varepsilon > 0$ such that $V_\varepsilon(x) \subseteq \mathring{\mathcal{C}} \subseteq \mathcal{C}$. Of course, this means that $\mathcal{C}$ would contain an open interval. This contradiction shows that $\mathcal{C}$ has empty interior. $\square$

Remark 1.1. Note that $0 = \inf \mathcal{C}$ and $1 = \sup \mathcal{C}$.² Since $0, 1 \in \mathcal{C}$, we see that $\mathcal{C}$ contains both of its “endpoints”. But this does not imply that $\partial\mathcal{C} = \{0, 1\}$! Indeed, as we shall see $\partial\mathcal{C}$ contains much more than the two endpoints 0 and 1 of $\mathcal{C}$.

We claim that $\partial\mathcal{C} = \mathcal{C}$. Since $\mathcal{C}$ is closed, the inclusion $\partial\mathcal{C} \subseteq \mathcal{C}$ is immediate. Conversely, let $x \in \mathcal{C}$ be given and fix $\varepsilon > 0$. Clearly, $V_\varepsilon(x) \cap \mathcal{C} \neq \emptyset$ since $x \in \mathcal{C}$. However, because $\mathcal{C}$ does not contain any open interval, we have $V_\varepsilon(x) \not\subseteq \mathcal{C}$. This implies that

$$V_\varepsilon(x) \cap \mathcal{C} \neq \emptyset \quad \text{and} \quad V_\varepsilon(x) \cap \mathcal{C}^c \neq \emptyset.$$

Since $\varepsilon > 0$ was arbitrary, it follows that $x \in \partial\mathcal{C}$. Thus, $\mathcal{C} \subseteq \partial\mathcal{C}$ and $\partial\mathcal{C} = \mathcal{C}$. In particular, $\partial\mathcal{C}$ is uncountable. Therefore, the Cantor set is an example of a closed set with boundary points that are **not** endpoints. In fact, every point in $\mathcal{C}$ is a boundary point of $\mathcal{C}$.

Definition 2. A subset X of $\mathbb{R}$ is said to be *totally disconnected* if, for any $x, y \in X$ with $x < y$, there exists $z \in \mathbb{R} \setminus X$ such that $x < z < y$.

¹A set $A \subseteq \mathbb{R}$ is said to be nowhere dense if the interior of $\overline{A}$ is empty.

²This is because $\mathcal{C} \subseteq [0, 1]$ and $0, 1 \in \mathcal{C}$.

One example of a totally disconnected set is $\mathbb{Q}$. Indeed, by the density of the irrationals, for any $x < y$ in $\mathbb{Q}$ one can always find an irrational number z with $z \in (x, y)$.

Proposition 3. *The Cantor set $\mathfrak{C}$ is totally disconnected.*

Proof. Let $x, y \in \mathfrak{C}$ be given and assume that $x < y$. We must show that there exists some $z \notin \mathfrak{C}$ with $x < z < y$. Let $\varepsilon := y - x > 0$ be the distance between x and y . Since $x, y \in \mathfrak{C}$, these two points must belong to every A_n . On the other hand, every subinterval making up A_n has length equal to $\frac{1}{3^n}$. Since

$$\inf_{n \geq 1} \frac{1}{3^n} = 0,$$

we can find N so large that $\frac{1}{3^N} < \varepsilon$. Since $x, y \in A_N$ and $|x - y| = \varepsilon > \frac{1}{3^N}$, x and y cannot belong to the same subinterval of A_N . Noticing that the subintervals of A_N are separated by open intervals, we can choose $z \notin A_N$ with $x < z < y$. Then, this z does not belong to $\mathfrak{C}$ and satisfies the condition of the proposition. $\square$

Definition 3. Let X be a non-empty subset of $\mathbb{R}$. We say that a point $x \in X$ is an *isolated point* of X if there exists $\varepsilon > 0$ such that $V_\varepsilon(x) \cap (X \setminus \{x\}) = \emptyset$.

In more familiar language, $x \in X$ is called an isolated point of X if there exists an ε -neighbourhood of x that does not contain points in X other than x .

Proposition 4. *The Cantor set $\mathfrak{C}$ has no isolated points.*

Proof. Fix $x \in \mathfrak{C}$ and let $\varepsilon > 0$ be given; we must show that $V_\varepsilon(x)$ intersects $\mathfrak{C}$ at a point other than x . Since $x \in \mathfrak{C}$, we must have $x \in A_n$ for all $n \geq 1$. Arguing as before, we can find $N \in \mathbb{N}$ so large that $3^{-N} < \varepsilon$. Consider this A_N , and recall that every subinterval of A_N has length equal to $3^{-N} < \varepsilon$. Let $I = [a, b]$ be the closed subinterval of A_N containing the point x . For any $y \in I$:

$$|x - y| \leq |b - a|. \text{ }^3$$

Since I has length $3^{-N} < \varepsilon$, it follows that

$$|x - y| \leq 3^{-N} < \varepsilon$$

³First note that $x \leq b$ and $y \geq a$. Therefore, $x - y \leq b - a$. Similarly, one can show that $y - x \leq b - a$ whence $-(b - a) \leq x - y$.

whence $y \in V_\varepsilon(x)$. Or, rather, $I = [a, b] \subseteq V_\varepsilon(x)$. Now, I is a subinterval from the construction of A_N whence $I \subseteq A_n$ for all $n \leq N$. On the other hand, when removing the “open middle third” in the construction, the end points of the intervals are never removed. Therefore the end points a, b of I will belong to A_n for all $n \in \mathbb{N}$. This simply means that

$$a, b \in \mathfrak{C} = \bigcap_{n \in \mathbb{N}_0} A_n.$$

Since $a, b \in V_\varepsilon(x)$, it follows that $V_\varepsilon(x)$ intersects $\mathfrak{C}$ at a point other than x . $\square$

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